



COMPANY OR INSTITUTION NAME

Department, Team or Reference Sector

Technical Report / Project Report

Project Reference, Client or Exam

EngTeX — Engineering L^AT_EX

Template

User Manual

Author:

Nome Cognome

Version:

Version 2.0

Date: July 7, 2026

Contents

1	Structure and General Settings	1
1.1	Document Version Management	1
1.2	Bibliography Usage (<code>biblatex</code>)	1
1.3	Main Module: <code>engtex.sty</code>	1
2	Boxes and Theorems Module: <code>engtex-boxes.sty</code>	3
2.1	What it does	3
2.2	How to Use (Code Example)	3
2.3	Full List of Features	3
2.4	Visual Result	3
3	Math Module: <code>engtex-math.sty</code>	4
3.1	What it does	4
3.2	How to Use (Code Example)	4
3.3	Full List of Features	4
3.4	Visual Result	4
4	Code Module: <code>engtex-code.sty</code>	6
4.1	What it does	6
4.2	How to Use (Code Example)	6
4.3	Full List of Features	6
4.4	Visual Result	6
5	Quick Reference and Cheat Sheet	8
5.1	Module Summary	8
5.2	Visual Cheat Sheet: Box Styles	10
	References	11

1. Structure and General Settings

The template is designed to provide an essential and modular environment for writing technical reports, theses, and documentation. Before diving into the details of the individual modules (math, boxes, etc.), this chapter covers the basic features included in the template and how to configure the main document.

1.1 Document Version Management

During the drafting of a document, it is crucial to keep track of the current version. The template allows showing a version identifier directly on the title page or in the footer. In the `main_ENG_Manual.tex` file (or equivalent), simply define the command:

```
1 \renewcommand{\versiondoc}{2.0}
```

The template automatically links this definition to the `\mostraversione` command (or its English alias `\viewversion`), which is called to print the version (for example, in `frontespizio.tex`). If you do not wish to display the version, just leave it empty or comment it out.

1.2 Bibliography Usage (biblatex)

The bibliography is managed automatically using `biblatex` with the `biber` backend. To insert sources:

1. Add your references in BibTeX format to the `layout/bibliografia.bib` file.
2. Cite sources in your text using the `\cite{bibtex_key}` command (e.g. `[1]` or `\parencite{paper_esempio}`).
3. Print the bibliography at the end of the document by inserting the `\printbibliography` command.

1.3 Main Module: `engtex.sty`

The base package `engtex.sty` is the core of the Lite template. It acts as an "Hub" to load all other modules and manages the language and style settings for the entire document. This allows you to selectively enable or disable template features: if you don't need to write code or draw TikZ diagrams in a project, you can simply avoid loading the corresponding module, speeding up compile times.

How to Use (Code Example)

The package is loaded in the preamble of the `main.tex` document. It uses a key-value interface for configuration.

```
1 \usepackage[  
2   language=english, % Document language: 'italian' or 'english'  
3   modules={math, boxes, code, tikz} % Modules to load in Lite version  
4 ]{package/engtex}
```

Available Options

- **language:** Automatically sets translations for algorithms, theorems, and layout labels (italian or english).
- **modules:** A comma-separated list of submodules to activate.

2. Boxes and Theorems Module: `engtex-boxes.sty`

This module manages the aesthetics of all information boxes and mathematical environments (theorems, definitions, etc.).

2.1 What it does

It allows you to call structured elements semantically, automatically applying a professional layout design.

2.2 How to Use (Code Example)

The syntax is: `\nomemacro{Title}{Text}` for elements requiring a title (e.g. definitions, theorems, examples), or `\nomemacro{Text}` for notes and warnings.

```
1 \definition{Causality}{A system is causal if...}
2
3 \note{Ensure that the sampling frequency satisfies the Nyquist-Shannon theorem.}
```

2.3 Full List of Features

Below are all the information and mathematical environments available in the Lite template:

- **Definition:** `\definition{Title}{Text}` | **IT:** `\definizione`
- **Note:** `\note{Text}` (No title) | **IT:** `\nota`
- **Warning:** `\warning{Text}` | **IT:** `\attenzione`
- **Example:** `\example{Title}{Text}` | **IT:** `\esempio`

2.4 Visual Result

Definition: Causality

A system is causal if the output at time t depends only on the values of the input at times prior to or equal to t .

Note: *This is an example of a sidebar note block.*

3. Math Module: engtex-math.sty

This module extends standard L^AT_EX capabilities by providing optimized macros for frequent math and engineering formulas, reducing code verbosity.

3.1 What it does

It simplifies typing complex operators like derivatives, integrals, transforms (Fourier, Laplace, Z), and vector operators (gradient, divergence, curl), ensuring excellent spacing and typographical formatting.

3.2 How to Use (Code Example)

```
1 The Laplace transform of a derivative is:
2 \begin{equation}
3   \laplace{\der{f(t)}{t}} = s \laplace{f(t)} - f(0)
4 \end{equation}
5
6 Calculating curl and divergence:
7 \begin{align}
8   \curl \mathbf{E} &= - \frac{\partial \mathbf{B}}{\partial t} \\
9   \div \mathbf{B} &= 0
10 \end{align}
```

3.3 Full List of Features

Below are all the math macros available:

- **Equation Systems:** `\system{ x = 1 \y = 2 }`
- **First Derivative:** `\der{f}{t}`
- **Fourier Transform:** `\fourier{x(t)}`
- **Laplace Transform:** `\laplace{f(t)}`
- **Phasor:** `\phasor{V}{\phi}` or `\fasore`
- **Negation:** `\notlog{A}`
- **Number Sets:** `\RR`, `\NN`, `\ZZ`, `\CC`
- **Norm and Absolute Value:** `\norm{v}`, `\abs{x}`
- **Vector Operators:** `\grad` (∇), `\div` ($\nabla \cdot$), `\curl` ($\nabla \times$), `\lap` (∇^2)
- **3x3 Matrix:** `\matrixthree{1}...{9}`

3.4 Visual Result

The Laplace transform of a derivative is:

$$\mathcal{L} \left\{ \frac{df(t)}{dt} \right\} = s \mathcal{L} \{ f(t) \} - f(0) \quad (3.1)$$

Calculating curl and divergence:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{3.2}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{3.3}$$

4. Code Module: `engtex-code.sty`

Inserting code snippets in a technical report requires excellent readability and correct syntax highlighting. The `engtex-code` module leverages `listings` to provide pre-configured code blocks.

4.1 What it does

It allows you to insert source code formatted with professional color schemes. Optimized pre-configured languages include: VHDL (Vivado style colors), C/C++, and Java. It also supports inline syntax for referencing variables or functions in plain text.

4.2 How to Use (Code Example)

An example of C code (use `\cinline{int sum = 0;}` in the text):

```
\begin{lstlisting}[style=c]
int calculate_mean(int values[], int len) {
    // Returns the arithmetic mean
    int sum = 0;
    for(int i=0; i<len; i++) sum += values[i];
    return sum / len;
}
\end{lstlisting}
```

4.3 Full List of Features

Below are all the supported languages and their inline commands:

- **VHDL:** `\begin{lstlisting}[style=vhdl]` | Compact: `[style=vhdlcompact]` | Inline: `\vhdlinline{...}`
- **C:** `\begin{lstlisting}[style=c]` | Inline: `\cinline{...}`
- **C++:** `\begin{lstlisting}[style=cpp]` | Inline: `\cppinline{...}`
- **Java:** `\begin{lstlisting}[style=java]` | Inline: `\javainline{...}`

4.4 Visual Result

An example of C code (use `int sum = 0;` in the text):

```
1 int calculate_mean(int values[], int len) {
2     // Returns the arithmetic mean
3     int sum = 0;
4     for(int i=0; i<len; i++) sum += values[i];
5     return sum / len;
6 }
```



```
entity D_FlipFlop is
  port ( clk, d : in std_logic; q : out std_logic );
end entity;
```

Listing 4.1: *D Flip Flop declaration in VHDL*

5. Quick Reference and Cheat Sheet

5.1 Module Summary

Module: `math`

Code	Function	Use Case
<code>\system{x=1 \y=2}</code>	$\begin{cases} x = 1 \\ y = 2 \end{cases}$	Equation systems
<code>\der{f}{t}</code>	$\frac{df}{dt}$	Ordinary derivatives
<code>\fourier{x(t)}</code>	$\mathcal{F}\{x(t)\}$	Signal Fourier transforms
<code>\laplace{f(t)}</code>	$\mathcal{L}\{f(t)\}$	Control Laplace transforms
<code>\fasore{V}{\phi}</code>	$V\angle\phi$	AC circuits calculations
<code>\notlog{A \cdot B}</code>	$\overline{A \cdot B}$	Digital electronics
<code>\RR, \NN, \ZZ, \CC</code>	$\mathbb{R}, \mathbb{N}, \mathbb{Z}, \mathbb{C}$	Common number sets
<code>\norm{v}, \abs{x}</code>	$\ v\ , x $	Algebra & calculus
<code>\grad f, \dive \mathbf{v}, \curl \mathbf{v}</code>	$\nabla f, \nabla \cdot \mathbf{v}, \nabla \times \mathbf{v}$	Vector calculus
<code>\lap f</code>	$\nabla^2 f$	Laplacian wave eq.
<code>\matrixthree{1}..{9}</code>	$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$	Matrices insertion

Module: `boxes`

Code	Function	Use Case
<code>\definition / \definizione</code>	Definition Box	Core concept
<code>\note / \nota</code>	Note Box	Sidebar annotations
<code>\warning / \attenzione</code>	Warning Box	Cautions and mistakes
<code>\example / \esempio</code>	Example Box	Practical examples

Module: code

Code	Function	Use Case
<code>\begin{lstlisting} [style=vhdl]</code>	VHDL Block	FPGA coding
<code>\begin{lstlisting} [style=c]</code>	C Block	Embedded systems
<code>\begin{lstlisting} [style=cpp]</code>	C++ Block	Object-oriented code
<code>\begin{lstlisting} [style=java]</code>	Java Block	Software design
<code>\begin{lstlisting} [style=lang compact]</code>	Compact code	Long listings (e.g. <code>vhdlcompact</code>)
<code>\lang inline{code}</code>	Inline code	Variables in text

5.2 Visual Cheat Sheet: Box Styles

Below is the visual appearance of all boxes provided by the template.

Definition: Definition

Example of a definition box.

Note: *Example of a note box.*

Warning: Example of a warning box.

Example 5.1 – Example

Example of a numbered example box.

References

- [1] A. Turing, “Computing machinery and intelligence,” *Mind*, vol. 59, no. 236, pp. 433–460, 1950.